

# RITESH PANDEY

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🐙 Github

📄 Leetcode

📁 Portfolio

## PROFILE SUMMARY

Final-year B.Tech CSE With at IIIT Kottayam with hands-on experience building AI-powered automation systems using LLMs, RAG pipelines, and agentic frameworks. Skilled in Python, LangChain, vector databases (FAISS, ChromaDB), and full-stack development. Passionate about applying AI to solve real-world customer challenges and building intelligent systems that reduce manual effort and improve user experience. Experienced in stakeholder coordination and technical event management through leadership roles.

## EDUCATION

**Indian Institute of Information Technology, Kottayam**

**2023 – 2027**

*B.Tech CSE with Specialisation in Cyber Security*

*Kottayam, Kerala*

### Relevant Coursework

Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Probability and Statistics, Database Management Systems, Cloud Computing, DSA, Object-Oriented Programming, Computer Networks.

## EXPERIENCE

**BetaLabs – Technical Club, IIIT Kottayam**

**Oct 2024 – Present**

*Lead, PR & Outreach (formerly Sub-Lead: Oct 2024 – Nov 2025)*

*Kottayam, Kerala*

- Leading cross-functional outreach and coordination with students, sponsors, and external stakeholders for technical events, ensuring smooth end-to-end execution.
- Managing event communications, incident handling, and stakeholder escalations – developing strong customer-facing communication and problem-resolution skills.
- Collaborating with team members to plan, organise, and deliver technical workshops and hackathons, driving engagement and community growth.

## PROJECTS

**DocRAG – Intelligent Document Assistant** 🔗 | React, Django, ChromaDB, LangChain, Groq API, JWT, Tailwind CSS | Jun – Jul 2026

- Built an AI-powered document assistant enabling users to instantly query and extract answers from uploaded PDFs, reducing manual document search effort through intelligent automation.
- Implemented Hybrid Retrieval (MMR + BM25) with ChromaDB and LangChain, achieving high contextual accuracy for automated document Q&A.
- Designed the system as a reusable AI agent architecture with multi-turn conversational support, persistent chat history, and autonomous document understanding.
- Developed secure REST APIs using Django REST Framework and JWT authentication, supporting chat management, PDF uploads, and user-specific document retrieval.

**SuggestBot – AI Query Automation Agent** 🔗 | Streamlit, LangChain, FAISS, SQLite, Ollama (LLaMA 2), Gemini API, RapidFuzz | Jun – Apr 2026

- Built an autonomous NL-to-SQL agent that converts natural language questions into executable database queries, automating data retrieval for non-technical users.
- Designed a multi-step agentic workflow: query understanding → fuzzy matching → SQL generation → execution → natural language response – reducing manual data lookups.
- Built a context-aware RAG chatbot with Gemini API, LangChain, and SQLite conversational memory for intelligent follow-up handling.
- Reduced query failure rate by 40% using typo-tolerant fuzzy matching (RapidFuzz) and embedding-based semantic retrieval using FAISS.

**HopAlong – Campus Ride Sharing** 🔗 | React.js, Node.js, PostgreSQL, Centrifuge, GeoApiFy, TypeScript | Jun – Apr 2025

- Developed a full-stack ride-sharing platform using React.js and Node.js with secure user authentication and real-time ride workflows.

- Designed REST APIs for ride matching, booking, and real-time communication between frontend and backend services.
- Implemented PostgreSQL schemas and modular backend logic for ride matching, cost splitting, and ride history management.
- Integrated GeoApiFy for location-based functionality and Centrifuge for real-time messaging between users.

## TECHNICAL SKILLS

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**Languages:** Python, SQL, C++, Java, JavaScript, C

**AI / LLM:** LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search, Semantic Search, FAISS, ChromaDB, T5, Gemini API, Groq API, Ollama (LLaMA 2), NL-to-SQL Systems.

**Machine Learning:** Supervised Learning, Unsupervised Learning, Regression, Classification, Feature Engineering, Model Training, Model Evaluation, Hyperparameter Tuning, PyTorch, TensorFlow, Hugging Face Transformers, Natural Language Processing.

**Data Science:** Pandas, NumPy, Scikit-learn, Data Cleaning, Data Preprocessing, EDA, Data Visualisation, Statistical Analysis, Cross-validation, Matplotlib, Regex.

**Frontend:** React.js, Streamlit, TypeScript, Tailwind CSS, Framer Motion.

**Backend:** FastAPI, Django, REST APIs, Async APIs, Node.js, Express.js, JWT Authentication

**Databases:** PostgreSQL, MySQL, SQLite, MongoDB, Supabase

**Cloud & Tools:** AWS (EC2, S3, IAM, Lambda, RDS, VPC), Docker, Git/GitHub, Linux, Postman, Cursor, GitHub Copilot, Excel.

## ACHIEVEMENTS

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- Supervised Machine Learning: Regression and Classification – DeepLearning.AI & Stanford University (Coursera)
- Completed a 5-day Faculty Development Program (FDP) on Responsible NLP (RNLP) at IIIT Kottayam.
- Solved 600+ DSA problems across LeetCode (229), GeeksforGeeks (293), and Coding Ninjas (114).